

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY
Third Semester of MSc(BT/BC/MI) Examination November 2019
MS812 Biostatistics

Date:08/11/2019, Friday

Time:01:30 p.m. to 01:40 p.m.

Maximum Marks:10

MCQ

Important Instructions:

- Tick the correct answer/s and it should be written in question paper itself.
 - Use of non-programmable calculator is allowed.
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Q - I Choose the correct answer for the following questions.

1. The standard deviation of 4 observations 5,5,5,5 is_____.
(i) 25 (ii) 5 (iii) 0 (iv) 5/3
2. Heterozygous black guinea pigs (Bb) are crossed among themselves and three offsprings are observed, if p is the probability of black offspring then $p =$ _____.
(i) 1/2 (ii) 3/4 (iii) 1/4 (iv) 3/8
3. A random variable X has poisson distribution with parameter λ , and $P(X = 0) = 0.135335$ then $\lambda =$ _____.
(i) 1 (ii) 7.4 (iii) 1.2 (iv) 2
4. Suppose $X_1, X_2, \dots, X_n$ are observations in random sample drawn from population with parameters mean μ and standard deviation σ respectively. Then the standard error of estimate of μ is _____.
(i) $\sigma / \sqrt{n}$ (ii) σ / n (iii) $\sigma / \bar{X}$ (iv) σ

5. The relation ship between central moment m_2 and the raw moments m'_1, m'_2 about arbitrary point A, is _____.

(i) $m_2 = m'_2 - A$

(iii) $m_2 = m'^2_2 - m'_1$

(ii) $m_2 = m'_2 - m'^2_1$

(iv) $m_2 = m'_2 - A^2$

For questions 6. to 8.: A researcher obtained regression line of the age (years) and systolic blood pressure of 20 apparently healthy adults ($s_x = 17.51$ is the standard deviation of age values, $s_y = 8.06$ is the standard deviation of blood pressure values, and $cov(x, y) = 136.47$ is the covariance between age and blood pressure) as $BP = 0.445Age + 112.315$

6. The correlation coefficient between the variables Age and Blood pressure is _____.

(i) 0.0828

(ii) 0.4603

(iii) 0.9670

(iv) 0.1283

7. The estimated value of blood pressure of individual of age 50 years is _____.

(i) 122.315

(ii) 112.315

(iii) 107.865

(iv) 134.565

8. The slope of the regression line from the diagram is _____.

(i) 0.445

(ii) 112.315

(iii) 17.51

(iv) 8.06

For questions 9. and 10.: Three analysts each made six determinations of the paracetamol content of the same batch of tablets. Consider the ANOVA Table (1)

Table 1: ANOVA TABLE

Source of Variation	SS	df	MSS	Fratio	Fcrit
Between Groups	0.2997	a	b	—	—
Within Groups	0.2799	c	—		
Total	0.5796	17			

9. From the Table (1), Value of $a =$ _____.

(i) 3

(iii) 17

(ii) 2

(iv) 15

10. From the Table (1), Value of $b =$ _____.

(i) 0.0999

(iii) 1.0707

(ii) 0.5994

(iv) 0.1499

End of MCQ

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Date:08/11/2019, Friday

Time:01:40 p.m. to 03:30 p.m.

Maximum Marks:25

Instructions:

- Section I and II must be attempted in SINGLE ANSWER SHEET.
- Make suitable assumptions and draw neat figures wherever required.
- Use of non-programmable calculator is allowed.
- Show necessary calculations.

SECTION I

Q - II Answer the following questions as directed. [10]

1. Define the following terms. [02]
 - (a) Arithmetic Mean and one property of arithmetic mean.
 - (b) Types of regression model.
2. The Table (2) shows two samples of cholesterol measurements, each on the same person, using different measurement techniques.

Table 2: cholesterol measurement

Autoanalyzer Method:	177	193	195	209	226
Microenzymatic Method:	192	197	200	202	209

Compute for both methods,

- (a) mean deviation about median [04]
- (b) standard deviation [04]

SECTION II

Q - III Answer the following questions as directed (**Attempt any three**). [15]

1. A nine -laboratory cooperative study was performed to evaluate quality control for susceptibility tests with 30- μ g netilmicin disks. The mean zone diameters for each of the nine laboratories are given in Table (3)

Table 3: Mean zone diameters with 30- μ g netilmicin disks tested in nine separate laboratories (control strain: P. aeruginosa)

Laboratory	A	B	C	D	E	F	G	H	I
Different media	20.1	18.4	16.8	21.7	20.1	20.3	22.8	19.9	19.3

Construct the 95% confidence interval for the mean zone diameter across laboratories for the type of control strain. You are given

$$t_{0.025,8} = 2.306, t_{0.025,9} = 2.262$$

2. In pharmacologic research a variety of clinical chemistry measurements are routinely monitored closely for evidence of side effects of the medication under study. Suppose typical blood-glucose levels are normally distributed, with mean = 90 mg/dL and standard deviation = 38 mg/dL. What percentage of values will fall in the range 65 - 120 mg/dL? You are given $P(Z \leq 0.6579) = 0.7447, P(Z \leq 0.7895) = 0.7851$
3. Black hair in the guinea pig is dominant to white hair. In families of 5 offspring where both parents are heterozygous black, with what frequency would we expect to find (a) 3 whites and 2 blacks (b) all whites (c) 1 white 4 blacks?
4. Forced expiratory volume (FEV) is a standard measure of pulmonary function. It is proposed that FEV is related to both age and height. Table (4) presents the mean FEV in liters for each of six 4-cm height groups. Find the best-fitting regression line $FEV = b_0 + b_1 \text{Height}$

Table 4: Mean FEV by height group for boys ages 10-15

Height(cm)	158	162	166	170	174	178
Mean FEV(L)	2.7	3.0	3.1	3.4	3.8	3.9

5. Consider a laboratory experiment in which three different chemicals (A, B and C) were added to cultures of a virus growing within Petri dishes. The purpose of the experiment was to compare the effects of the chemicals on the growth of the virus, measured by the colony diameter. The experiment was arranged as a completely randomised design with a total of 15 units. The observed data are given in Table (5)

Table 5: Diameter (mm) of each affected area

A	17.1	14.9	23.6	16.0	18.7
B	24.3	21.7	22.3	26.4	23.2
C	14.3	14.5	17.1	13.2	15.6

Test the hypothesis $H_0 : \mu_1 = \mu_2 = \mu_3$ against $H_1 : \mu_1 \neq \mu_2 \neq \mu_3$. Where μ_1 , μ_2 and μ_3 are true mean diameter of affected area. Use 5% level of significance. You are given $F_{2,12,0.05} = 3.885$.

6. It is suspected that an acid-base titrimetric method has a significant indicator error and thus tends to give results with a positive systematic error (i.e. positive bias). To test this an exactly 0.1 M solution of acid is used to titrate 25.00 ml of an exactly 0.1 M solution of alkali, with the following results (ml):

Table 6: Measurements of experiment (ml)

25.06	25.18	24.87	25.51	25.34	25.41
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If μ represent the true mean (ml) of experiment values, then write appropriate hypothesis H_0 against H_1 . Test the hypothesis H_0 at 5% significance level. You are given $t_{0.05,5} = 2.015$